

What Happens Next?

Video Action Recognition on Something-Something v2

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Abstract

We present our approach to the *What Happens Next?* Kaggle challenge: 33-class video action recognition on a curated subset of Something-Something v2 (SSv2) [4], where only the first $\approx 40\%$ of each clip is provided. Track A (Closed World) achieves 44.03% top-1 accuracy on the official validation set using a from-scratch ensemble of Temporal Shift Module [7] (TSM-ResNet18) and VideoFormer-Lite models. Track B (Open World) reaches **0.6586 Kaggle top-1** through a 14-model uniform ensemble built on V-JEPA 2 [2] and VideoMAE [9] backbones, with progressive unfreezing, layer-wise learning-rate decay (LLRD), and window-capped source-frame re-extraction. A self-discovered data-leakage event (a discarded 0.81 submission) is documented alongside the integrity audit that triggered a full pipeline rebuild.

1 Problem Description

Task. The challenge asks models to classify 4-frame JPEG clips into one of 33 motion-defined action classes (e.g. *Pouring something into something*, *Pulling something from left to right*). Crucially, each clip contains only the *anticipatory* portion of the original SSv2 video: the shipped `frame_003` sits at a median fraction of 0.40 of the source clip, so the action’s outcome is deliberately withheld. This design forces models to reason about motion trajectories rather than final-state configurations.

Dataset. The provided split contains 44,993 training clips, 6,745 validation clips, and 6,913 test clips across 33 classes (numeric prefixes 000–032; class 027 is absent from the training set). Class sizes are heavily imbalanced: the largest class (009 — *Moving something up*) has 3,170 training examples while the smallest (026 — *Spilling something next to something*) has only 162, a ratio of $19.6\times$. The KL divergence between the training and validation class distributions is 0.088, confirming that the splits are representative.

Evaluation. Submissions are evaluated by Top-1 accuracy on a public+private Kaggle leaderboard. The Top-5 metric is reported internally. A provided baseline achieves approximately 0.55 on the private leaderboard. The second-place public reference score is 0.74.

Two-track setup. Track A (Closed World) prohibits pre-trained models and external data; models are trained from scratch on the provided 4-frame clips. Track B (Open World) permits pretrained models and external datasets provided that privileged information (i.e. future frames or the ground-truth SSv2 labels for the test set) is not used.

Contributions. Andrea Signoretti designed and implemented the full Track A pipeline (Section 2.1), including TSM-ResNet, VideoFormer-Lite, regularization warm-up, and the ensemble/TTA framework. Florian Guillaume designed and implemented the full Track B pipeline (Section 2.2), including source-frame re-extraction with window-capping, the V-JEPA wrapper with progressive unfreezing and LLRD, pseudo-labelling, the cross-preprocessing ensemble, and the data-integrity audit.

2 Methods

2.1 Track A — Closed World

The Track A pipeline evolved through six architectural stages, guided by ablation results on an internal training-split validation.

2.1.1 Architecture family

CNN Baseline / CNN-LSTM. A ResNet-18 backbone with temporal average-pooling served as the starting point. Adding a unidirectional LSTM over per-frame features yielded only marginal improvement due to instability of backpropagation through time on short 4-frame sequences.

VideoFormer-Lite (VFL). Per-frame ResNet-18 features are passed to a [CLS] token together with learnable temporal positional embeddings and processed by N pre-norm Transformer encoder blocks (multi-head self-attention + MLP). The final [CLS] representation is classified. The *ultra* variant uses $N=4$ layers and a tuned learning rate, reaching 53.37% on the internal split.

TSM-ResNet. The Temporal Shift Module [7] wraps every residual block of ResNet-18, shifting $1/4$ of channels backward and $1/4$ forward in time at zero parameter cost. TSM was designed specifically for SSv2-style motion-defined labels, making it a strong prior for this task.

2.1.2 Critical frame-count fix

The single largest accuracy gain in Track A (+4.4 pp) came from aligning the model’s `num_frames` parameter with the true clip length. Clips contain exactly 4 real JPEG frames; setting `num_frames > 4` triggers frame interpolation, which corrupts the temporal-shift signal that TSM relies on. Setting `num_frames = 4` removed this corruption and brought `tsm_ultra_v2` to 57.73% internal validation.

2.1.3 Regularization recipe

All Track A models share a common regularization stack whose most critical element is a *warm-up delay* on MixUp [12] and CutMix [11]: these techniques are disabled for the first 10 epochs so the backbone can form basic per-frame representations before label-mixing is applied. This single change produced a +36 ppjump on VFL (9.45% $\rightarrow$ 45.78%). The full recipe also includes label smoothing ($\epsilon=0.1$) [8], RandAugment, random erasing [13], stochastic depth (DropPath 0.1), dropout 0.2 before the classifier, and AdamW with cosine annealing. Focal loss ($\gamma=2$) is used in `tsm_ultra_v2` to focus gradient on confusable class pairs.

CutMix probability is deliberately kept low ($p=0.25$) because a cut patch spans all 4 frames and frequently occludes the hand/object motion that defines the label.

Domain-aware augmentation is essential: horizontal flip is disabled during training because SSv2 direction-encoded classes (e.g. *Pulling left to right*) change their ground-truth label under a flip.

2.1.4 Final Track A ensemble

The final submission fuses three checkpoints via log-softmax late fusion with 10-crop spatial TTA (5 spatial crops $\times$ 2 flips): `tsm_ultra_v2` (57.73% internal), `tsm_ultra_v2_rotating` (rotating-fold variant trained on all data), and `video_former_lite_ultra_rotating`. The ensemble reaches **44.03% on the official `val_dir`**.

2.2 Track B — Open World

2.2.1 Backbone selection

Top-performing teams on SSv2 benchmarks consistently employ video transformers pretrained on large unlabelled corpora. We selected two families:

VideoMAE [9] uses masked autoencoders on video: 90% of video patches are masked and the encoder must reconstruct them, learning rich spatio-temporal representations. We use the HuggingFace checkpoints pretrained on Kinetics-400

(`videomae-base-finetuned-kinetics`, `videomae-large-finetuned-kinetics`).

V-JEPA 2 [2] is a joint-embedding predictive architecture that predicts representations of masked video regions in *feature space* rather than pixel space. This avoids the reconstruction of visually uninformative regions (backgrounds, textures) and focuses capacity on semantically relevant motion. We use the self-supervised checkpoint `vjepa2-vitl-fpc64-256`, which was pretrained on unlabelled video without any SSv2 labels — a critical requirement for integrity (Section 3.5).

2.2.2 V-JEPA fine-tuning strategy

V-JEPA was not present in the original codebase; we implemented the full fine-tuning wrapper, including the frame-extraction pipeline and the LLRD helper, from scratch. The strategy proceeds in three stages:

Stage 0 — Frozen attentive probe. A randomly initialised linear head is trained for 6 epochs with all encoder weights frozen. The EMA validation accuracy on the training split reaches 0.672.

Stage 1 — Partial unfreezing with LLRD. The top-6 transformer blocks are unfrozen. Layer-wise learning-rate decay [1, 3, 5] is applied: the head receives the base learning rate η ; each lower block ℓ receives $\eta \cdot \lambda^\ell$ with $\lambda=0.75$. Warm-starting from Stage 0 weights, 4 epochs bring EMA validation to 0.705.

Stage 2 — Full encoder fine-tuning with LLRD. All encoder layers are unfrozen; LR decay is extended to the full depth. Warm-starting from Stage 1, 4 more epochs bring EMA validation to 0.728. On the official `val_dir` evaluation, the final V-JEPA-ft checkpoint achieves 0.6411 top-1.

The progressive unfreezing strategy follows the intuition that randomly initialised classification heads can corrupt lower-layer representations if allowed to propagate large gradients from the first epoch. Freezing the backbone initially stabilises the head, and graduated unfreezing preserves the pretrained feature geometry.

2.2.3 Source-frame re-extraction

V-JEPA benefits from 16 input frames, whereas only 4 frames per clip are provided. The folder names in the dataset encode SSv2 video IDs, so the full source WebM files (220,847 videos, $\approx$ 19 GB) are accessible.

We implemented `extract_ssv2_frames.py` to re-extract 16 real frames per clip in a *window-capped* manner: (1) the shipped `frame_003` is matched against low-resolution source frames using pixel-level mean absolute error to locate the end of the provided temporal window; (2) 16 frames are sampled uniformly within `[0, window_end]`; (3) the search is hard-bounded to the

first 50% of the source clip, so the withheld action outcome ($>60\%$) is unreachable by construction. Empirically, the resulting median window fraction is 0.39 and the maximum is 0.50.

2.2.4 Pseudo-labelling

After Stage 2 fine-tuning, the V-JEPA-ft top-3 snapshot ensemble with 10-view TTA generated soft labels for the 6,913 test clips. Clips with maximum softmax probability ≥ 0.85 (100% coverage at this threshold) were incorporated as pseudo-labelled examples. The model was then continue-finetuned on $\text{train} \cup \text{pseudo-test}$ with full encoder + LLRD for 4 more epochs, following the noisy-student recipe [6, 10]. The resulting V-JEPA-pseudo achieves 0.6519 `val_dir` and 0.6410 Kaggle; the widened val/Kaggle gap (Section 3.2.4) indicated overfitting to the validation distribution, and pseudo-labelling was not iterated further.

2.2.5 Honest cross-preprocessing ensemble

The ensemble aggregates 14 model checkpoints from three architectural families, each fine-tuned with its own preprocessing pipeline:

- **V-JEPA** (6 snapshots): 16-frame, 256 px, window-capped;
- **VideoMAE-Base K400** (3 snapshots): 4-frame, 224 px, shipped frames;
- **VideoMAE-Large K400** (3 snapshots): 4-frame, 224 px, shipped frames;
- **TSM-ResNet50** (2 snapshots): 4-frame, 224 px, shipped frames.

Inference runs each model on its own preprocessing chain, producing per-model softmax tensors that are cached to disk. `honest_ensemble.py` then combines them by uniform averaging of the raw softmax scores.

The key empirical finding is that *uniform averaging of diverse honest models outperforms learned or manual weighting*. When gradient descent was used to fit per-model weights on validation NLL, the optimiser concentrated $\approx 82\%$ of the weight on the two strongest VideoMAE snapshots and effectively zeroed the K400 and TSM members. Yet uniform weighting with those same weak members included scored higher, because the architecturally distinct models produce uncorrelated errors that average out.

2.2.6 Infrastructure

Training ran on a single RTX 4000 Ada (21 GB VRAM). Three infrastructure challenges required original engineering: (1) a silent NFS quota (30 GB) was producing 0-byte checkpoints until `fsync`-durable atomic saves turned the

Table 1: Summary of Track A and Track B results. † Best honest submission.

Trk	Method	val	Kaggle
A	TSM-ResNet18 (single)	38.25%	—
A	VFL-Ultra (single)	32.11%	—
A	3-model ens. + TTA	44.03%	—
B	V-JEPA-ft standalone	64.11%	0.6361
B	Ens. v1 (9-uniform)	—	0.6418
B	V-JEPA-pseudo	65.19%	0.6410
B	12-uniform	—	0.6546
B	14-uniform†	65.19%	0.6586

silent failure into an honest `OSError`, prompting migration of all checkpoints to a local NVMe (`/Data`, 561 GB); (2) a top- K snapshot rename race condition caused snapshots to overwrite each other during training — fixed by a two-phase staging rename; (3) cross-architecture ensembling was silently loading Large-model weights into a Base graph because both save `model_name="videomae"` — fixed by rebuilding each model from its own stored config at inference.

3 Results

3.1 Overall performance

[Table 1: Track summary results — val accuracy and Kaggle score by track and submission type]

Table 1 reports the headline results for both tracks.

3.2 Track B ablations

3.2.1 Honest Kaggle progression

[Figure 1: Bar chart of Kaggle scores over submissions, red bar for leaky/discarded result, blue bars for clean submissions]

Table 2 shows the complete Track B Kaggle history. The 21 pp gap between the discarded leaky run (0.81) and the first honest baseline (0.6361) quantifies what the data leakage was purchasing. Every subsequent gain was earned through legitimate architectural improvements or ensemble diversity.

3.2.2 Ensemble size and weighting

[Figure 2: Line chart of Kaggle score vs. number of ensemble members, annotated to show that dropping weak models hurt]

[Table 3: Uniform vs. manual weighting for same model sets]

Table 3 isolates the effect of ensemble size and weighting strategy on the same model collection.

The V-JEPA-only ensemble (row 1) *regresses* from the 9-model uniform ensemble despite V-JEPA’s higher individ-

Table 2: Track B Kaggle submission history. **Red**: discarded leaky run. **Green**: honest submissions. Δ is relative to V-JEPA-ft (first honest baseline).

Submission	Kaggle	Δ
Leaky V-JEPA (discarded)	0.81	—
V-JEPA-ft standalone (clean)	0.6361	—
Ensemble v1 (9-uniform)	0.6418	+0.6 pp
V-JEPA-pseudo standalone	0.6410	+0.5 pp
V-JEPA-only (6 snapshots)	0.6410	regression
V-JEPA-heavy (3:1 manual)	0.6499	+1.4 pp
12-uniform (+ pseudo)	0.6546	+1.9 pp
14-uniform (+ Large)	0.6586	+2.3 pp

Table 3: Ensemble strategy ablation. All rows use the same 12 checkpoints except where noted.

Strategy	n	Kaggle
V-JEPA-only, uniform	6	0.6410
9-uniform (+ K400 + TSM)	9	0.6418
V-JEPA-heavy (3:1 weight)	12	0.6499
12-uniform	12	0.6546
14-uniform	14	0.6586

ual accuracy, confirming that the weaker VideoMAE-K400 and TSM models contribute uncorrelated errors that the ensemble corrects. The manual V-JEPA-heavy weighting (row 3) falls between uniform-9 and uniform-12: domain knowledge about model quality helps less than simply adding diverse members.

3.2.3 Progressive unfreezing

[Figure 3: Multi-line training curve (EMA val accuracy vs. epoch) with stages in different colours, showing warm-start jumps between stages]

Figure 3 shows the V-JEPA training trajectory. The transition from the frozen probe to Stage 1 (top-6 unfrozen + LLRD) is characterised by a warm-start jump: Stage 1 epoch 1 (0.676) already exceeds Stage 0’s final epoch (0.672), and the curve climbs to 0.705 by epoch 4. Stage 2 repeats the pattern. The warm-start jump is a direct consequence of LLRD preserving low-layer representations while the higher learning rate on the head adapts the classification boundary.

3.2.4 Calibration

[Figure 4: Three-bar grouped chart (train-split EMA val, val_dir, Kaggle) for V-JEPA-ft and V-JEPA-pseudo]

Table 4 quantifies the calibration gap between the internal validation proxy and the true Kaggle score.

Table 4: Calibration: train-split EMA val, official val_dir, and Kaggle, for two V-JEPA checkpoints. The pseudo-label run’s val_dir→Kaggle gap doubles relative to the clean fine-tune.

Model	Train-split val	val_dir	Kaggle
V-JEPA-ft	0.7281	0.6411	0.6361
V-JEPA-pseudo	0.7415	0.6519	0.6410

Table 5: Track A leave-one-out ablation on val_dir (no TTA, 3-model weighted ensemble).

Configuration	Top-1	Δ
Full ensemble (3 models)	40.43%	—
– tsm.ultra.v2	37.26%	–3.17 pp
– tsm.ultra	39.56%	–0.87 pp
– video_former_lite.ultra	39.82%	–0.61 pp

The val_dir→Kaggle gap is 0.005 for V-JEPA-ft and 0.011 for V-JEPA-pseudo — a doubling that warned us against further pseudo-label iterations. The train-split → Kaggle gap (≈ 0.09) reflects that the internal train/val split was drawn from the same distribution, while the Kaggle test set is independently sampled.

3.3 Track A analysis

3.3.1 Ensemble component contributions

[Table 4: Leave-one-out ablation for Track A 3-model ensemble]

Table 5 shows the Track A leave-one-out ablation on the full val_dir (6,746 clips). All three components make a positive contribution; removing tsm.ultra.v2 alone costs more than the other two combined, confirming it as the dominant member.

3.3.2 Per-family accuracy (Track B)

[Figure 5: Horizontal bar chart of single-model val_dir top-1 accuracy by backbone family, colour-coded, with the 11-model uniform ensemble shown as the top bar]

Table 6 shows single-family val_dir top-1 accuracy versus the 11-model uniform ensemble. No individual family reaches the ensemble score, and the TSM family — despite its low standalone accuracy (0.34) — contributes to the ensemble’s improvement because its errors are the least correlated with those of any transformer-based family (inter-family agreement 0.37–0.40; Table 7).

Table 7 quantifies prediction diversity. TSM is the most complementary member: its pairwise agreement with every transformer family is below 0.40, compared to 0.58 for two VideoMAE variants. The intra-family agreement for V-JEPA (0.85) explains why a V-JEPA-only ensemble

Table 6: Per-family `val_dir` top-1 accuracy (Track B, 3 snapshots per family, single-view) and the 11-model uniform ensemble.

Family	Snapshots	val top-1
VideoMAE-Base K400 4f	3	0.5576
VideoMAE-Large K400 4f	2	0.5945
VideoMAE-Base SSv2 4f	3	0.6047
TSM ResNet-50 4f	3	0.3364
Uniform ensemble (11 models)	11	0.6222

Table 7: Pairwise inter-family prediction agreement on the test set. Lower agreement implies more complementary errors.

Family A	Family B	Agreement
V-JEPA-pseudo	TSM	0.370
V-JEPA-ft	TSM	0.372
K400-Base	TSM	0.375
V-JEPA-ft	K400-Base	0.582
V-JEPA-ft	V-JEPA-pseudo	0.854

underperforms a diverse one of equal size.

3.4 Failure analysis

[Figure 6: 33×33 confusion matrix heatmap for the Track B 11-model ensemble on `val_dir`]

3.4.1 Dominant error patterns

Three confusion patterns dominate both tracks and reflect the design intent of the SSv2 benchmark:

Real vs. pretended actions. The worst single confusion (Track B) is *Picking something up* (class 011) being predicted as *Pretending to pick something up* (class 014): 56 errors out of 199 validation clips, an 11.6% top-1 F1 for class 011. The two classes share an identical early-clip trajectory; the disambiguating cue (whether the object actually rises) falls in the withheld future. This is precisely the anticipation reasoning that the task is designed to probe. Focal loss ($\gamma=2$) in Track A partially addresses this but does not resolve it.

Temporal direction confusion. *Folding* $\leftrightarrow$ *Unfolding* (24–29 bidirectional errors) share identical spatial appearance; with 4 frames the temporal ordering is fragile.

Static sink class. *Holding something* attracts many misclassified clips from motion classes (*Moving something down*: 30 errors; *Opening*: 28), acting as the network’s default prediction when no decisive motion is detected in the provided frames.

3.4.2 Per-class analysis

The 5 best-predicted Track A classes ($F1 > 0.54$) are all single-arc motions: *Pulling from left to right*, *Pouring*, *Moving closer*, *Uncovering*, *Folding*. The 5 worst ($F1 < 0.20$) are *Picking up*, *Pretending to put*, *Pretending to throw*, *Spilling*, and *Showing to camera* — classes where early-clip evidence is ambiguous or intentionally misleading.

The correlation between training-set class size and per-class accuracy is weak ($r=0.24$), confirming that the difficulty stems from semantic ambiguity rather than sample scarcity.

3.4.3 The top-5 recovery gap

The top-5 accuracy on `val_dir` is 0.882, compared to 0.622 top-1, a gap of 0.260. This gap is largest for the hardest classes: *Pretending to put something into something* recovers from 11.8% top-1 to 79.4% top-5 (+67.6 pp), and *Picking something up* from 11.6% to 72.9% (+61.3 pp). The model assigns high probability to the correct class but ranks a visually similar “pretend” variant first. This suggests the model understands the action family but cannot reliably distinguish real from pretended execution without observing the outcome.

3.5 Data-integrity story

[Figure 7: Histogram of `frame_003` position (fraction of source video) for sampled clips, with dashed line at 0.40 and shaded region >0.50]

An early extraction run sampled frames uniformly over the full source clip (0–100%). Training with V-JEPA on these frames yielded 0.81 Kaggle — 7 pp above the public second-place. An audit revealed two concurrent leakage vectors: (1) frames from the withheld outcome portion ($>60\%$) were reachable, giving the model access to information the competition explicitly withholds; (2) the initial V-JEPA checkpoint used was finetuned on the SSv2 dataset with ground-truth labels, meaning the backbone had seen the test-set videos during its own pretraining.

Both runs were discarded and the entire pipeline was rebuilt as described in Section 2.2.3, switching to the self-supervised `vjepa2-vitl-fpc64-256` backbone and enforcing the 50% hard bound on extraction. We consider the 21 pp honest/leaky accuracy difference a direct measurement of the combined leakage effect.

4 Conclusion

We have presented a two-track solution to video action anticipation on SSv2. Track A demonstrates that carefully engineered from-scratch models — with domain-appropriate regularization and correct temporal sampling — can reach competitive accuracy (44.03%) on a notoriously

hard benchmark. Track B shows that progressive unfreezing with LLRD enables effective fine-tuning of large video transformers on a constrained GPU budget, and that *ensemble diversity across architectural families consistently outperforms single-family ensembling or learned weighting schemes*.

Failure analysis reveals a fundamental limit: the dataset’s design intent is to distinguish real from pretended actions and directional from reverse motions — disambiguations that require observing the withheld outcome. Improving on these classes likely requires either explicit trajectory prediction heads or iterative pseudo-label refinement with stronger confidence calibration.

Future work. The most promising next steps are: (1) including V-JEPA-Giant once inference-time memory constraints are addressed; (2) targeted class-weight rebalancing for the *holding* and *pretending* sinks; (3) a second round of pseudo-labelling with temperature-scaled confidence thresholds to avoid the val-distribution over-fitting observed after the first round; and (4) 17-model ensemble with a newly fine-tuned VideoMAE-Large 16-frame window-capped checkpoint currently in training.

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Figures and Tables

Figure 1 — Honest Kaggle progression. Vertical bar chart of Kaggle scores over all Track B submissions. The leaky bar (0.81) is shown in red with a strike-through label “DISCARDED”. All honest bars are shown in blue/green. A horizontal dashed line marks the 0.74 second-place reference. Data source: Table 2.

Figure 2 — Ensemble size vs. Kaggle score. Line + scatter plot. x -axis: number of models in the uniform ensemble; y -axis: Kaggle top-1. An annotation arrow at $n=6$ marks the V-JEPA-only regression: dropping the K400 and TSM members despite their lower individual accuracy *hurts* the ensemble. Data source: Table 3.

Figure 3 — V-JEPA progressive-unfreezing training curves. Multi-line plot of EMA validation accuracy vs. epoch on the training split. Stage 0 (frozen probe), Stage 1 (top-6 unfrozen + LLRD), and Stage 2 (full encoder + LLRD) are plotted in distinct colours. The warm-start jump at each stage transition is annotated. Data:

Stage	Epoch	EMA val acc
Frozen probe	1	0.5755
Frozen probe	6	0.6725
Stage 1 (top-6 + LLRD)	1	0.6764
Stage 1 (top-6 + LLRD)	4	0.7052
Stage 2 (full + LLRD)	1	0.7015
Stage 2 (full + LLRD)	4	0.7281
Pseudo retrain	1	0.7180
Pseudo retrain	4	0.7415

Figure 4 — Calibration comparison. Grouped bar chart with three bars per model: train-split EMA val, `val_dir`, Kaggle. Data source: Table 4.

Figure 5 — Per-family `val_dir` accuracy. Horizontal bar chart sorted by accuracy, colour-coded by family (V-JEPA, VideoMAE-Base SSv2, VideoMAE-Large, VideoMAE-Base K400, TSM). Data source: Table 6.

Figure 6 — Confusion matrix. 33×33 heatmap of the 11-model Track B ensemble on `val_dir`. Colour scale: white = 0, dark blue = max. The strongest off-diagonal clusters correspond to: *Picking up* $\rightarrow$ *Pretending to pick up* (col 014 in row 011), *Folding* $\leftrightarrow$ *Unfolding* (rows/cols 003, 032), *Moving down/up* $\rightarrow$ *Holding* (col 005). Full 33×33 CSV data available in supplementary material.

Figure 7 — Frame position histogram. Histogram of `frame_003`’s fractional position in the source clip, sampled from 14 training clips. A vertical dashed line at 0.40 marks the median end of the provided window. The shaded

region 0.50–1.00 is labelled “withheld future — data leakage if extracted”. An overlay shows the corrected extraction’s window-end distribution (median 0.39, max 0.50).